BookNest – Where the stories Nestle

Team Information

Team ID: LTVIP2026TMIDS81146
Team Size: 4

Team Members

Team Member : Supriya Perisetti

Role: Project Coordinator
Responsibilities:

- Oversees overall project planning and execution
- Manages GitHub repository and version control
- Coordinates frontend and backend integration
- Ensures timely completion of project milestones

• Team Leader : Sai Krishna

Role: Backend Developer
Responsibilities:

- Builds RESTful APIs using Node.js and Express.js
- Manages authentication and server-side logic
- Handles routing and middleware integration
- Ensures secure and efficient backend operations

Team Member : Durga Prasanth Vandana

Role: Frontend Developer
Responsibilities:

- Develops React-based user interface
- Designs reusable components
- Implements page routing using React Router

- Handles API integration and user interactions

Team Member : Mekala Srinivas

Role: Database Administrator
Responsibilities:

- Designs and manages MongoDB schemas
- Performs CRUD operations
- Maintains Users, Books, and Orders collections
- Ensures data consistency and security

1. Introduction

Project Title: BookNest – Where Stories Nest

BookNest is a modern digital bookstore application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). It is designed to transform the traditional book-buying experience into a seamless and engaging online platform. The application combines powerful backend technologies with an intuitive and responsive user interface to provide book lovers with a convenient way to explore, discover, and purchase books.

In today's fast-paced digital world, readers often struggle to find time to visit physical bookstores. BookNest addresses this challenge by offering a centralized online platform where users can browse a vast collection of books across various genres, authors, and categories from the comfort of their homes. The platform ensures a smooth browsing experience, secure authentication, efficient book search, and a simplified purchasing process.

The backend of the application is powered by MongoDB, which provides a scalable and efficient database system for storing book details, user information, and order records. Express.js serves as the web application framework, enabling the development of robust and responsive APIs. Node.js ensures high-performance server-side operations with non-blocking architecture, resulting in a seamless user experience.

At the heart of BookNest lies React.js, a dynamic JavaScript library used to create an interactive and visually appealing frontend. The responsive design ensures that users can access the application smoothly across desktops, tablets, and smartphones, delivering a consistent experience on all devices.

BookNest aims to bridge the gap between traditional bookstores and modern digital solutions by redefining how readers connect with literature. It offers a convenient, efficient,

and enjoyable way to explore books and complete purchases online, making reading more accessible in the digital era.

2. Project Overview

Purpose

The purpose of the BookNest project is to develop a full-stack web application that provides a modern and user-friendly online bookstore experience. The platform is designed to allow users to explore, search, and purchase books conveniently from anywhere at any time.

BookNest aims to eliminate the limitations of traditional bookstores by offering a digital solution where users can easily discover new releases, revisit classic literature, and manage their purchases efficiently. The system ensures smooth navigation, secure user authentication, and a seamless checkout process.

The primary objectives of the project are:

- To enable users to browse a comprehensive collection of books online.
- To provide advanced search and filtering options based on genre, author, price, and popularity.
- To allow users to securely register, log in, and manage their accounts.
- To implement a cart system for adding and managing selected books.
- To ensure secure purchase processing and order confirmation.
- To maintain order history for users to track their past purchases

Features

For Users:

✓ **Sign Up / Log In** – Users can create an account and securely log in to access personalized features.

✓ **Book Listings** – Displays a comprehensive list of available books with details such as title, author, genre, description, price, and availability.

✓ **Search & Filter** – Users can search books and filter results based on genre, author, and pricing.

✓ **Book Selection** – Allows users to select books and view detailed information before purchasing.

✓ **Add to Cart** – Users can add books to their shopping cart and manage quantities.

✓ **Purchase Process** – Secure checkout process for completing book purchases.

✓ **Order Confirmation** – Generates confirmation after successful purchase with order details.

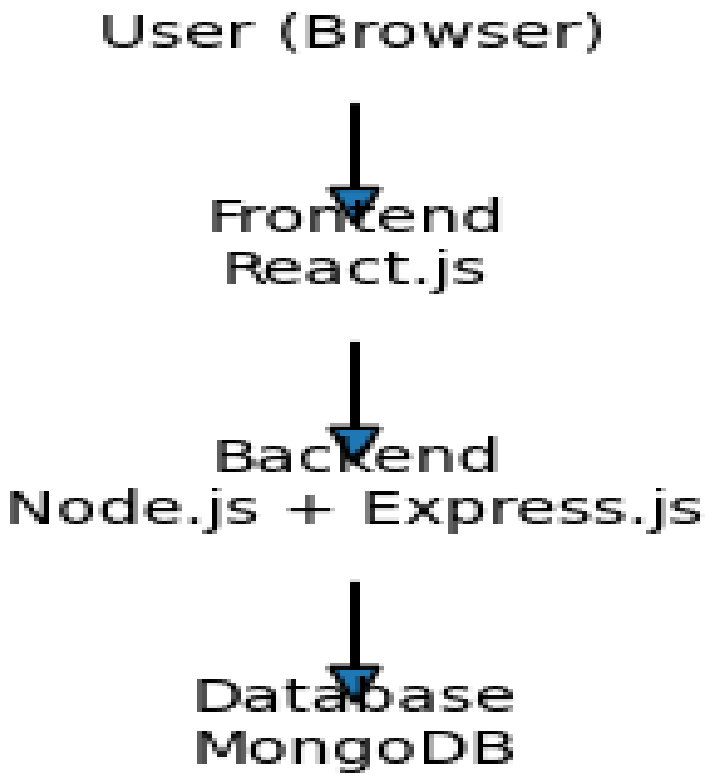
✓ **Order History** – Users can view their previous orders and transaction details.

3. Architecture

Frontend: React.js handles UI components and routing.

Backend: Node.js and Express.js manage APIs and business logic.

Database: MongoDB stores user, book, and order data.



4. Setup Instructions

Prerequisites

Before running the BookNest project, ensure the following software is installed on your system:

- **Node.js (v16 or above)** – To run frontend and backend
- **npm (Node Package Manager)** – Installed along with Node.js
- **MongoDB** – Local database (MongoDB Compass recommended)
- **Git** – To clone the repository
- **VS Code** – Recommended code editor

Installation Steps

Clone the Project

```
git clone https://github.com/supriya549/BookNest.git

cd BookNest
```

Backend Setup

1. Navigate to the server folder

```
cd server
```

2. Install dependencies:

```
npm install
```

3. Configure Environment Variables

Create a .env file inside the **server** folder and add:

```
PORT=5000
MONGO_URI=mongodb://localhost:27017/booknest
JWT_SECRET=your_secret_key
```

4. Start Backend Server:

```
node server.js
```

Frontend Setup

Open a new terminal window.

1. Navigate to client folder:

cd client

2. Install dependencies:

npm install

3. Start React Application:

npm start

Frontend will run at:

http://localhost:3000

Start MongoDB

If using MongoDB Compass:

- Open MongoDB Compass
- Connect to mongodb://localhost:27017
- Create a database named: booknest

If using terminal:

mongod

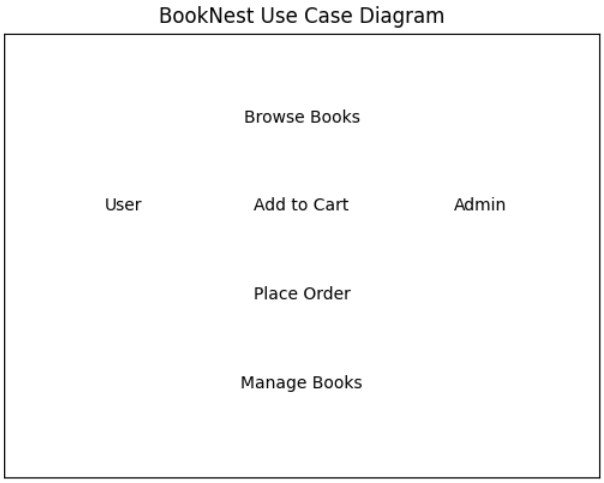
6. API Documentation

- POST /api/register – Register new user
- POST /api/login – Login user
- GET /api/books – Retrieve all books
- POST /api/orders – Place new order
- GET /api/orders/:userId – Retrieve user orders

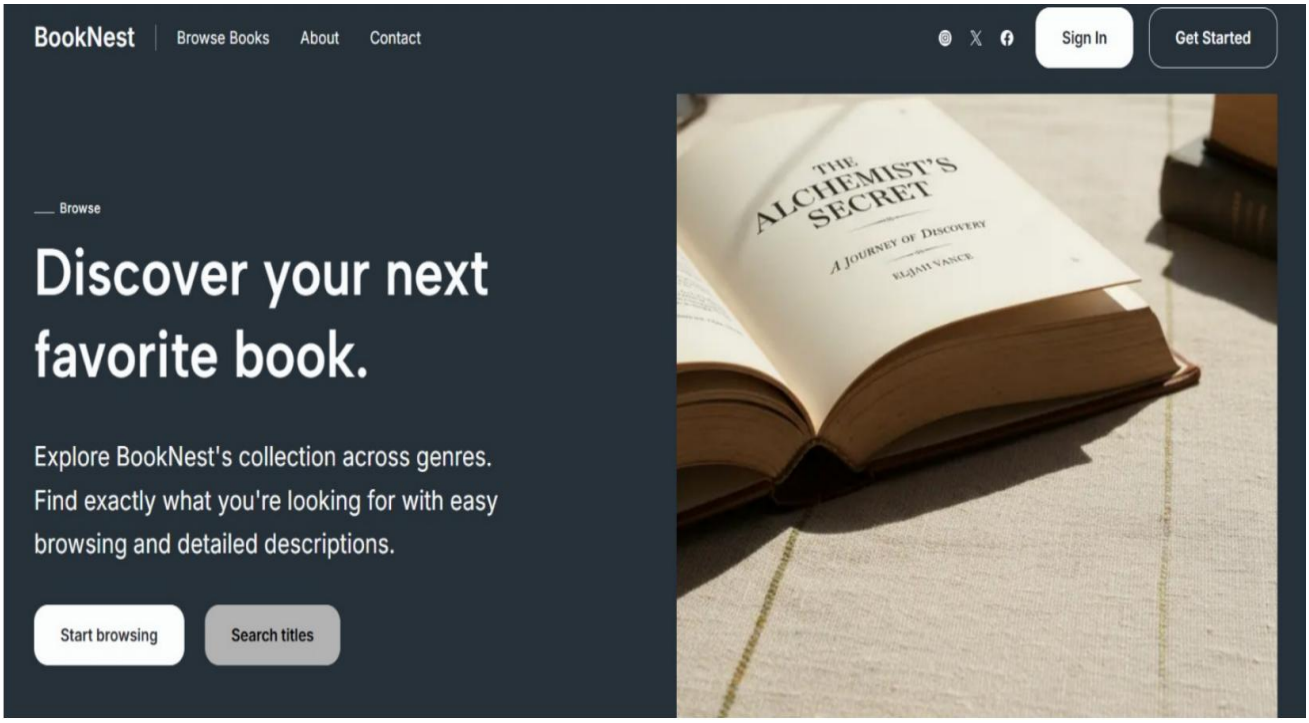
7. Authentication

Currently uses email and password authentication. Future enhancement includes implementing JWT-based authentication for secure sessions.

8. User Interface



Screenshots:



Fiction

Immerse yourself in captivating novels, literary works, and contemporary stories.

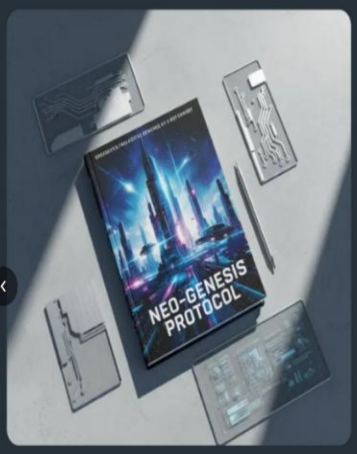
Non-Fiction

Learn from thought-provoking memoirs, biographies, and educational explorations.

Mystery & Thriller

Unravel suspenseful plots and gripping page-turners that keep you guessing.

contemporary stories. educational explorations. keep you guessing.



Science Fiction

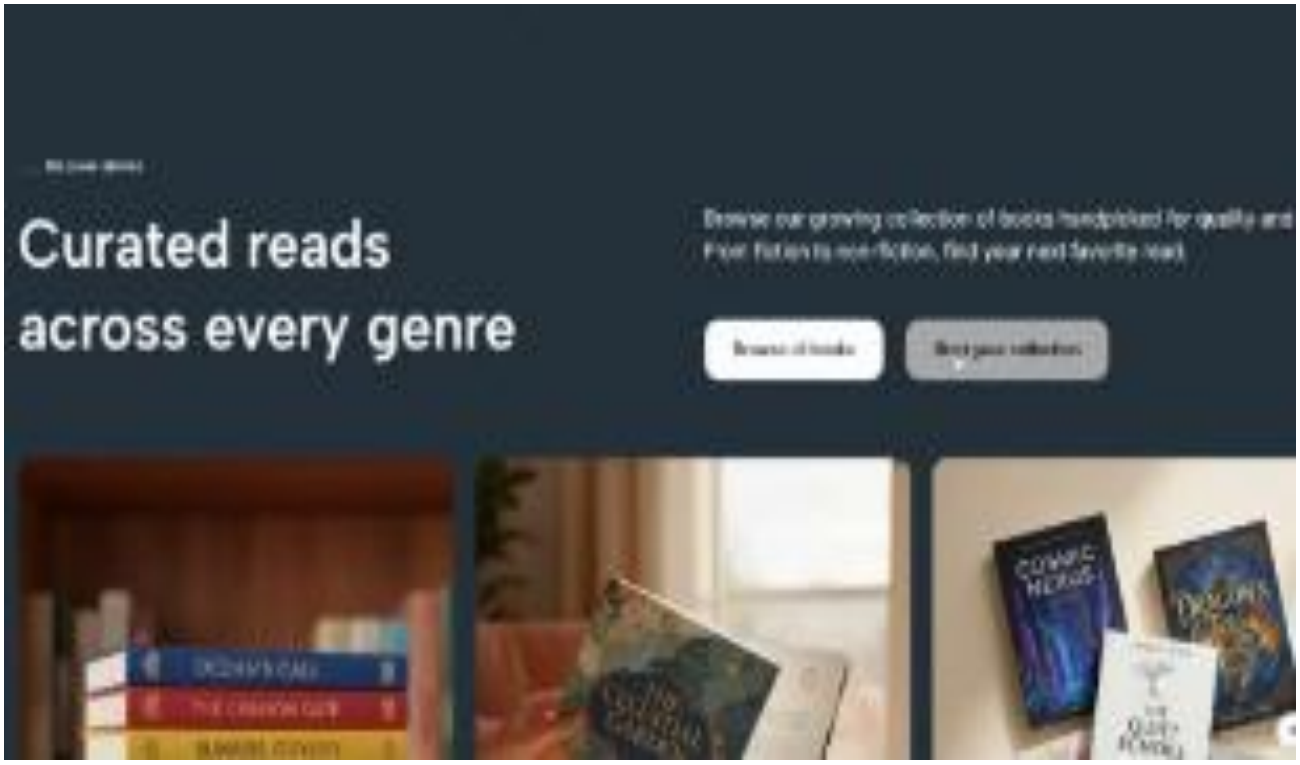
Venture into futuristic worlds, space exploration, and imaginative universes.

Romance

Enjoy heartfelt tales and emotional journeys about love and relationships.

Young Adult

Connect with coming-of-age stories and adventures written for modern readers.



9. Testing

Manual testing was performed for registration, login, book browsing, cart management, and order placement. Postman was used to test backend APIs.

11. Future Enhancements

- Implement JWT Authentication
- Add Payment Gateway Integration
- Develop Admin Dashboard
- Implement Recommendation System
- Add Automated Testing

GITHUB LINK : https://github.com/supriya549

VIDEO DEMO:
https://drive.google.com/file/d/1lJ3OnhfFfRPF29_BAo3XtYEak1wPxI6N/view?usp=drivesdk